Machine Learning Integration Strategy

Machine learning is central to Stripe's strategy for fraud detection and customer experience personalization. This document describes the process of integrating machine learning models, from feature extraction to production monitoring.

1. Feature Extraction

Features are the model's input data, stored in the `machine_learning_features` class of the NoSQL database. They are extracted in real time from three sources. The first is the OLTP database, containing the amount, currency, location, device type, and customer history. The second is the OLAP database, containing the customer's analytical history and the average fraud rate per country. The final source is the NoSQL database, containing user behavior data. Fafka enables the real-time transmission of transaction data to MongoDB to feed these features. This data is normalized and cleaned before being sent to the model. Each machine_learning_feature document is enriched with calculated features. These include the number of the customer's transactions in the last 24 hours, their usual spending amount, their usual country of residence, and the use of a new device.

2. Model Deployment

The models are trained separately on historical OLAP data such as fact_fraud and fact_transaction, which provide the target data to confirm, for example, fraud. Once trained, the model is deployed to production via an API. Stripe can then call it in real time for each transaction using a fraud score that constitutes the model's prediction. Multiple versions of the model can coexist to test their performance. They are all versioned in MLflow, making them traceable and reproducible. In case of a problem, a rollback to previous versions is therefore possible quickly.

3. Real-Time Fraud Detection

Fraud detection is the primary use case for models at Stripe. The model needs to be able to predict its fraud score as quickly as the transaction itself, so as not to slow it down. The most important features for the model are the detection of unusual amounts, new countries, multiple simultaneous transactions, or suspicious IP addresses. If fraud is detected, an alert is sent via Kafka to the `fraud_alerts` topic. The results are then stored in the OLAP `fact_fraud` to improve future training.

4. Customer Personalization

A second model analyzes customer behavior to personalize their experience. The input data comes from user_interactions in MongoDB. This includes pages visited, session durations, buttons clicked, and transaction history. The model can predict which payment methods will be offered first and which sales offers to display. These predictions are stored in MongoDB and used in real time for the Stripe application.

5. Predictive Analytics

A third model predicts future trends based on historical OLAP data. It anticipates which sellers will leave the Stripe platform and traffic spikes. These models run in batch mode nightly via Airflow and do not require real-time prediction. The results enrich OLAP dimensions such as dim_merchant and dim_customer.

6. Model Monitoring and Updates

Model performance is continuously monitored using metrics. Accuracy allows us to see how many fraud detections were correct. Recall allows you to see how many frauds were detected out of all those that occurred. The F1 score shows the balance between accuracy and Recall, and latency shows the model's prediction time. These metrics are stored in MongoDB and can be visualized in a dashboard. An alert is sent if performance falls below a defined threshold.
Monitoring also aims to detect model drift. This can occur for two different reasons: Data drift happens when the distribution of input data changes, for example, with the emergence of a new device to connect to the platform. Concept drift can occur when the relationship between features and fraud changes, for example, with the emergence of a new fraud technique.
To prevent model drift, the models are regularly trained on new data. Regarding the frequency of these training sessions, Stripe has enough data to retrain the models every month. Therefore, retraining must be automatically triggered if a drift is detected. New models are tested before being deployed in production. If the new models underperform in real-world conditions, a rollback is automatically performed.

Thus, three models coexist. They address fraud detection, customer experience personalization, and predictive analytics. Extracting features from the three systems guarantees rich and accurate predictions. Continuous monitoring ensures model reliability, and regular retraining allows adaptation to new fraud techniques.